

SK ASPAK ALI

Software Engineer / VoLTE/IMS Testing / Network Engineering

Bhubaneswar, India | +91 7655843136 | shaikhaspakali007@gmail.com

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PROFESSIONAL SUMMARY

Telecom Software Engineer with 24+ months of hands-on experience in VoLTE/IMS testing and validation at Nokia, supporting major carriers (Vodafone Idea, Bharti Airtel, Deutsche Telekom). Expert in protocol analysis (SIP, Diameter, DNS), network troubleshooting, and test automation, having rapidly mastered complex telecom architectures while delivering enterprise-grade solutions. Proficient in designing and executing complex test scenarios, identifying critical signaling issues, and ensuring network reliability across 3G/4G/5G deployments. Track record of delivering measurable results in high-pressure, multi-carrier environments.

KEY ACHIEVEMENTS

- Identified and resolved 45+ critical IMS signaling issues (SIP, Diameter, DNS) across 3 major carriers, achieving 82% first-time resolution rate and preventing 15+ potential production incidents.
- Executed and validated 520+ complex VoLTE/IMS test scenarios across 3G/4G/5G networks, reducing regression testing cycle time by 50% and maintaining 98% test quality accuracy.
- Diagnosed root causes of 28 critical network failures using advanced Wireshark trace analysis, enabling rapid deployments affecting 15M+ subscribers across 10+ telecom circles.
- Developed custom Python test validation scripts to automate Diameter AVP compliance checks, identifying 15+ compliance violations before production deployment.
- Mentored 4 junior engineers on IMS protocols and advanced Wireshark analysis, enabling them to independently resolve complex signaling issues within 8 weeks.
- Supported emergency call (911/112) scenario validation for DTAG, ensuring compliance with regulatory requirements across 4 network domains.

PROFESSIONAL EXPERIENCE

Software Engineer | Technoshrine Infosolution Pvt Ltd (Client: Deutsche Telekom, Germany - DTAG)

Apr 2026 – Present

- Executed 1cIMS Solution Validation Strategy across 28 end-to-end test scenarios (SBC, CFX, VoLTE, roaming, SMS over IP, emergency calls); identified 7 critical configuration issues, enabling core team to resolve routing misconfigurations and achieve carrier-grade interoperability.
- Performed root cause analysis on 35+ protocol traces (SIP, Diameter, DNS), diagnosing interoperability issues in 24 scenarios and resolving 18 pre-production bugs that would have impacted live traffic across 4 network domains.
- Validated SMS over IP routing consistency across 4 network domains, discovering and documenting 6 routing inconsistencies and ensuring 100% compliance with 3GPP standards and Deutsche Telekom carrier-specific requirements.
- Automated 28 IMS test cases using DTAG's GTAF (Test Automation Framework), reducing manual execution time by 50% and enabling faster regression validation cycles.

- Resolved 14 critical escalations in collaboration with cross-functional German teams (product managers, network architects), achieving 50% first-time resolution rate and accelerating resolution of critical escalations and reducing response time.
- Validated emergency call (911/112) scenarios ensuring 100% regulatory compliance across all 4 network domains, preventing potential public safety system failures.

Software Engineer | Technoshrine Infosolution Pvt Ltd (Client: Bharti Airtel India Ltd)

Apr 2025 – Apr 2026

- Executed end-to-end VoLTE and NLD (National Long Distance) testing across 10 telecom circles (Rajasthan, Gujarat, Maharashtra, Karnataka, Tamil Nadu, etc.), validating 185 test scenarios and identifying 42 network functionality issues affecting carrier-grade reliability.
- Analyzed 320+ Wireshark packet traces to troubleshoot SIP/Diameter signaling issues, identifying root causes in 28 scenarios and delivering 25 critical bug fixes to carrier engineering teams, reducing call setup failures by 50%
- Resolved critical interoperability issues between Bharti's HSS and third-party network elements (IMS core, SBC, gateways), reducing call setup failures by 60% and improving network reliability metrics.
- Documented comprehensive best practices playbook for VoLTE testing methodology and created internal knowledge base with 40+ troubleshooting guides, reducing onboarding time for new team members from 4 weeks to 2.5 weeks.
- Supported HSS redundancy and failover validation across 8 circles, enabling zero-downtime network expansion initiatives and ensuring business continuity for 8M+ active subscribers.
- Mentored 2 junior engineers on IMS call flows, Diameter protocol analysis, and Wireshark trace interpretation, enabling them to independently resolve complex signaling issues within 6 weeks (50% faster than standard 12-week training curve).

Software Engineer | Technoshrine Infosolution Pvt Ltd (Client: Vodafone Idea India Ltd)

July 2024 – Apr 2025

- Executed VoLTE UE registration testing and call setup validation across 18+ subscriber scenarios (voice calls, video calls, SMS-over-IP), achieving 100% test success rate and enabling immediate production deployment.
 - Conducted comprehensive core IMS functional testing for voice, video, and RCS messaging services across 3 IMS domains, validating HSS signaling integrity and identifying 12 critical service delivery issues.
 - Supported HSS capacity expansion project: installed, configured, and integrated 4 new HSS nodes, executed validation testing under 10M+ subscriber load simulation, and verified performance metrics (response time, throughput, latency) met SLA requirements.
 - Identified critical race condition in IMS call release flow using detailed Diameter trace analysis, preventing potential call completion failures that would have affected 100K+ daily call attempts in production.
 - Validated network performance metrics: call setup latency (reduced to <500ms), call completion rate (improved to 99.2%), and service availability (maintained 99.95% SLA compliance).
 - Designed and validated 15+ lab test scenarios for HSS failover and redundancy, enabling zero-downtime maintenance windows and achieving 100% service availability during carrier expansion.
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PROJECTS & TECHNICAL CONTRIBUTIONS

- **VoLTE Interoperability Testing Framework:** Built end-to-end test validation suite across Nokia, Ericsson, and 2 vendor-specific IMS implementations, uncovering 22 critical interoperability flaws. Drove vendor coordination on 18 accepted bug fixes, reducing interop validation timeline by 35%.
- **Wireshark Trace Analysis Tool:** Engineered Python-based bulk trace analysis tool automating Diameter AVP extraction, error code classification, and latency correlation across 500+ traces. Eliminated 70% of manual trace analysis overhead and improved issue identification accuracy to 96%.
- **IMS Network Health Dashboard:** Built real-time anomaly detection system integrating SIP/Diameter/DNS metrics with automated alerting. Flagged 45+ potential issues pre-production, preventing 12+ customer-impacting incidents and shifting team from reactive to predictive operations.

EDUCATION

B.Tech in Computer Science & Engineering — Trident Academy of Technology, Bhubaneswar (2020 – 2024) | CGPA: 8.3/10

TECHNICAL SKILLS

Category	Skills
Telecom Protocols & Testing	SIP (RFC 3261), Diameter (RFC 3588/6733), DNS, IMS architecture, VoLTE, NLD, HSS, S-CSCF, IMS signaling.
Tools & Platforms	Wireshark (advanced trace analysis), DTAG GTAF (test automation), SBC, CFX, Tester Studio, Nokia IMS simulators.
Programming & Automation	Python (test scripts, trace analysis), SQL, Linux, Bash scripting, test case design and execution.
Network & Cloud	TCP/IP networking fundamentals, AWS Cloud (hands-on labs), network simulation, Linux system administration.

CERTIFICATIONS & TRAINING

- L3 NTSP Test Automation Framework — Nokia Service Portal (2026)
- SVT Integration Engineer Certificate — Nokia Service Portal (2024)
- AWS Certified Cloud Practitioner — Amazon Web Services (2024)
- Full Stack Web Development — Udemy (Angela Yu, 2023)